Account Statement Running Balance

Difficulty: Beginner | Topic: Records and Files | Language: COBOL

1. Problem Statement

A statement generator applies a series of credits and debits to an opening balance. The program must process each transaction in order and display the running balance after every transaction.

2. Task

Read an opening balance and N transactions. Each transaction contains a type and an amount:

- C — Credit: add the transaction amount to the current balance.
- D — Debit: subtract the transaction amount only if the result will not be negative.
- If a debit would make the balance negative, reject it and retain the previous balance.

3. Input Format

- Line 1: Opening balance.
- Line 2: Number of transactions, N.
- Next N lines: Transaction type and amount, separated by whitespace.

4. Output Format

Print the resulting balance after each transaction. For a rejected debit or invalid transaction type, print the error marker in the form REJECTED: followed by the unchanged balance.

5. Constraints

Parameter	Constraint
Opening balance	0 ≤ opening balance ≤ 10,000,000
Number of transactions	1 ≤ N ≤ 1000
Transaction amounts	Non-negative integers

6. Sample Input

```
1000
3
C 200
D 500
D 900
```

7. Sample Output

```
1200
700
REJECTED: 700
```

9. COBOL Program

The following program uses the transaction input as two whitespace-separated values. The ACCEPT statements for transaction type and amount are written separately to match the provided code's line-based input style. If the judge supplies both values on one line, use the input-reading approach required by that environment.

```
IDENTIFICATION DIVISION.
PROGRAM-ID. STATEMENT-GENERATOR.

DATA DIVISION.
WORKING-STORAGE SECTION.
01 BALANCE-AMOUNT PIC 9(8) VALUE 0.
01 N              PIC 9(4) VALUE 0.
01 I              PIC 9(4) VALUE 0.
01 TRANS-TYPE     PIC X.
01 TRANS-AMOUNT   PIC 9(8) VALUE 0.
```

```
PROCEDURE DIVISION.
    ACCEPT BALANCE-AMOUNT.
    ACCEPT N.

    PERFORM VARYING I FROM 1 BY 1
        UNTIL I > N

        ACCEPT TRANS-TYPE
        ACCEPT TRANS-AMOUNT

        EVALUATE TRANS-TYPE
            WHEN "C"
                ADD TRANS-AMOUNT TO BALANCE-AMOUNT
                DISPLAY BALANCE-AMOUNT

            WHEN "D"
                IF TRANS-AMOUNT <= BALANCE-AMOUNT
                    SUBTRACT TRANS-AMOUNT FROM BALANCE-AMOUNT
                    DISPLAY BALANCE-AMOUNT
                ELSE
                    DISPLAY "REJECTED: " BALANCE-AMOUNT
                END-IF

            WHEN OTHER
                DISPLAY "REJECTED: " BALANCE-AMOUNT
        END-EVALUATE

    END-PERFORM.
    STOP RUN.
```

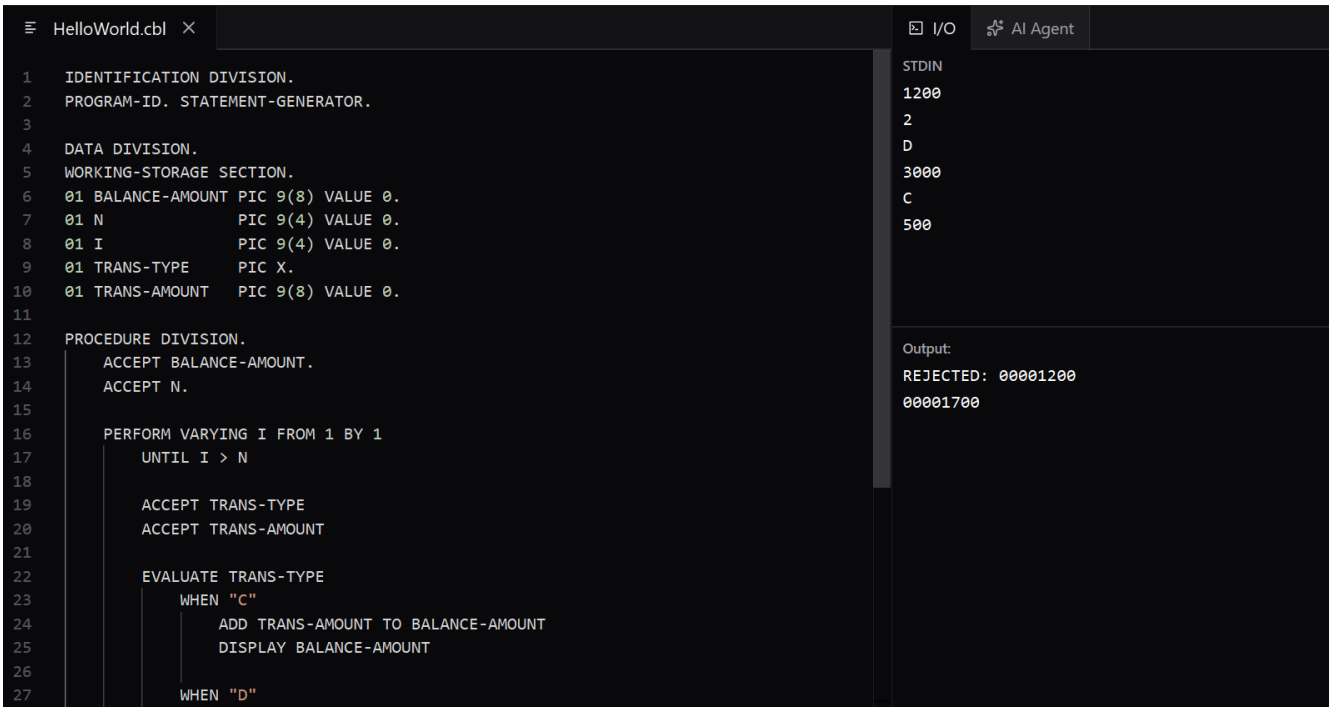


Fig 1. Program Execution and Output